

Patrick Tierney

(319) 330-5881 | ptierney13@gmail.com | Kirkland, WA 98033 | [linkedin.com/in/ptierney13](https://www.linkedin.com/in/ptierney13)

SUMMARY

Software engineer with 8 years of experience building developer platforms, observability systems, CI/CD infrastructure, and internal tooling at Microsoft. Specialized in developer productivity, operational reliability, telemetry systems, and distributed data processing supporting engineering organizations of 100+ developers and products serving millions of users. Experienced designing systems that improve engineering effectiveness through scalable tooling, safe abstractions, and operational visibility.

EXPERIENCE

Microsoft

Software Engineer II (2019 – 2025), **Software Engineer I** (2017 – 2019)

Platform Engineering & Developer Productivity

- Architected and led delivery of a unified testing platform used by 150+ engineers supporting Copilot Hub, standardizing local and CI validation workflows across Jest, Playwright, and Storybook-based visual testing
- Defined and implemented AI-assisted static analysis and PR-time validation enforcing accessibility and compliance testing requirements while reducing developer overhead
- Analyzed high-conflict CI workflows within Copilot Hub and introduced scheduling strategies that reduced average commit validation time from ~4 hours to ~40 minutes for heavily contended builds
- Co-designed and implemented an OpenTelemetry instrumentation framework for Microsoft Whiteboard, creating developer-safe telemetry abstractions that prevented privacy and compliance violations while supporting GCC, GCCH, and DoD cloud separation requirements

Data Pipelines, Observability & Information Systems

- Designed and implemented a low-latency telemetry processing pipeline in Azure that reduced rollout health visibility from hours to minutes, enabling rapid detection of production regressions during deployments
- Enhanced a telemetry query DSL used by 1,000+ Edge engineers, improving analytics, diagnostics, and rollout analysis capabilities for large-scale feature experimentation
- Developed automation that generated operational dashboards and rollout visibility systems directly from telemetry data for new feature deployments
- Implemented Microsoft Graph APIs and OAuth/JWT authorization flows for Whiteboard GDPR-oriented data access and deletion workflows, integrating Azure AD scopes and permissions

Reliability, Compliance & Engineering Operations

- Established frontend incident response and operational support practices that reduced engineering support workload by 70% and incident resolution times by 80%
- Built tooling and operational runbooks enabling customer support teams to resolve common production issues without direct engineering escalation
- Maintained 99.9% availability for Microsoft Whiteboard during major COVID-era educational scaling supporting millions of users globally
- Worked with GCC, GCCH, and DoD compliance constraints to ensure environment-specific telemetry and data separation requirements were enforced correctly

TECHNICAL SKILLS

Languages & Application Development: TypeScript, JavaScript, C#, .NET, SQL, React, Python, C++, R
Platform & Infrastructure: Azure Services, Azure DevOps, YAML CI/CD Pipelines, PowerShell, Distributed Systems
Observability, Testing & Reliability: OpenTelemetry, Jest, Playwright, OAuth/JWT Authentication, Distributed Tracing, Incident Response, Telemetry Systems

EDUCATION

Harvey Mudd College

Bachelor's Degree, Computer Science and Mathematics; Concentration in Economics

Claremont, CA

May 2016